

The effects of aerobic exercise on working memory in methamphetamine-dependent patients: Evidence from combined fNIRS and ERP

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ABSTRACT

Objective: To explore the effect of aerobic exercise on working memory according to exercise intensity in patients with methamphetamine (MA) dependence.

Design: Twelve-week randomized controlled trial.

Method: Forty-five MA-dependent patients were divided randomly into three groups participating in 12-week moderate- and high-intensity aerobic exercise interventions and no intervention (control), respectively. Working memory performance was assessed using a 2-back task, and a combined functional near-infrared spectral imaging–event-related potential measure was used to identify spatial and temporal components of the neurofunctional underpinnings of this performance.

Results: High-intensity exercise significantly enhanced the discernment aspect of working memory. It also decreased N1 and P2 amplitudes, and increased neural activity in the left dorsolateral prefrontal cortex.

Conclusions: The findings of this study suggest that high-intensity aerobic exercise can enhance working memory and neural activity in MA-dependent patients, and thus that it may serve as an effective therapeutic tool for these patients.

1. Introduction

A review of the current literature clearly shows that aerobic exercise positively influences the repair of cognitive function (Etnier, Drollette, & Slutsky, 2019; Piepmeyer & Etnier, 2015). This exercise-induced benefit has also been observed in methamphetamine (MA)-dependent patients, mainly through the enhancement of inhibitory control from behavioral and neural perspectives (Huang, Zheng, Gao, Hu, & Yuan, 2020; Wang, Zhou, & Chang, 2015; Wang, Zhu, Zhou, & Chang, 2017). Working memory, a fundamental component of cognitive function, has also been proven to play an important role in drug withdrawal; individuals with weak working memory are more likely to relapse (Chambers, Hugh, & Bellgrove, 2009). However, the effect of exercise on working memory among MA-dependent patients remains unknown.

MA dependence is increasingly recognized as a relapsing brain disorder and, in support of this view, marked structural changes in the striatal and prefrontal brain regions have been reported (Alizadehgoradel et al., 2020; Ersche et al., 2012). Existing studies show that aerobic exercise enhances prefrontal cortex (PFC) activity and improves working memory performance in patients with abnormal brain structure, such as children with attention-deficit/hyperactivity disorder (ADHD) (Chaddock-Heyman et al., 2013; Suarez-Manzano,

Ruiz-Ariza, Torre-Cruz, & Martínez-López, 2018) and post-stroke patients (Moriya, Aoki, & Sakatani, 2016). Meanwhile, neuroimaging studies suggest that the PFC is involved in working memory (Braver et al., 1997). Thus, aerobic exercise may have efficacious effects on working memory in MA-dependent patients, with changes in the PFC playing a crucial role, although direct evidence is needed to support this speculation.

Although the cognitive benefit of exercise is well known, it may vary according to exercise intensity. Numerous studies have shown that exercise-induced brain activity changes are modulated by exercise intensity (Shibuya, Sadamoto, Sato, Moriyama, & Iwadate, 2008), with greater intensity associated with greater cerebral oxygenation and blood volume in the brain (Gondoh et al., 2009). Current research shows that high-intensity aerobic exercise facilitates inhibitory control performance in MA-dependent patients (Wang, Zhou, Zhao, Wu, & Chang, 2016). Some researchers have reported significant increases in working memory performance and brain-derived neurotrophic factor (BDNF), insulin-like growth factor 1, and cortisol concentrations in healthy high-intensity exercise groups compared with those observed in low- and moderate-intensity exercise groups (Jeon & Ha, 2017). Similarly, Hwang et al. (2016) found that acute high-intensity aerobic exercise improved prefrontal-dependent cognitive performance and

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increased BDNF concentrations in healthy young adults. In addition, high-intensity exercise acutely increased prefrontal brain activity in patients with Parkinson's disease (Kelly et al., 2017). Jones (2015) reported that moderately intense aerobic exercise did not improve working memory in young children with ADHD. Currently, no study has investigated the relationship between exercise intensity and working memory benefits in MA-dependent patients. As such, we compared the effects of the three exercise intensities in this study in an attempt to further understand any dose-response relationship.

Most previous studies have used event-related potentials (ERPs) to measure dynamic processing in the brain cortex during cognitive task performance in MA-dependent patients. This analysis enables the detection of temporal components of working memory processing. For example, the N1 component is an index of discrimination processing in working memory (Vogel & Luck, 2000), and the P2 component is indicative of the initial stage of context updating related to working memory (Lenartowicz, Escobedo-Quiroz, & Cohen, 2010). However, spatial neural changes are difficult to measure using ERPs. To facilitate the observation of brain hemodynamic changes, we also employed functional near-infrared spectral imaging (fNIRS), a noninvasive neuroimaging method used widely to monitor PFC activity during cognitive task performance. Thus, we used a combined fNIRS-ERP measure to synchronously identify spatial and temporal components of the neuro-functional underpinnings of working memory processing. PFC activation and related ERP features were investigated.

The aim of this study was to determine the effects of aerobic exercise at various intensities on working memory among MA-dependent patients. We designed 12-week moderate- and high-intensity aerobic exercise interventions, and applied a combined fNIRS-ERP measure during participants' working memory task performance for assessment. We expected that 12 weeks of aerobic exercise would enhance working memory in MA-dependent patients, and that the effects of high-intensity exercise would be stronger than those of moderate-intensity exercise.

2. Material and methods

2.1. Participants

Forty-five participants were recruited from the Zhejiang Shiliping Drug Rehabilitation Bureau in China. All participants were aged 18–45 years and met the DSM-V criteria for MA-dependent patients, according to the Structured Diagnostic Interview. The exclusion criteria were 1) a history of psychosis, 2) physical or medical conditions that contraindicated exercise and 3) abstinent for 3 months.

All participants were divided randomly into three groups: moderate- and high-intensity exercise groups and a control group. The demographic characteristics of the groups are shown in Table 1. Participants in all three groups received routine care provided by staff at the rehabilitation center. This care included behavior management, nutritional support, physical examination, legal education, and discipline education.

Table 1
Subjects' demographic characteristics (M ± SE).

	Control group (N = 15)	High-intensity group (N = 15)	Moderate-intensity group (N = 15)
Age(year)	33.87 ± 1.98	32.73 ± 1.31	29.40 ± 1.19
Number of relapse(time/year)	1.67 ± 0.25	2.20 ± 0.24	1.87 ± 0.27
Duration of drug use(year)	7.33 ± 1.10	7.07 ± 0.61	5.87 ± 0.88
Weight(kg)	70.89 ± 1.93	70.71 ± 3.00	71.40 ± 2.76
Height(m)	1.71 ± 0.011	1.69 ± 0.015	1.71 ± 0.015
BMI	24.09 ± 0.64	24.54 ± 0.90	24.31 ± 0.90
Duration of education(year)	8.60 ± 0.50	10.00 ± 0.63	9.40 ± 0.27

Number of relapse means the mean relapse time per year before abstinence

No significant difference in any characteristic was detected among groups (all $p > 0.05$)

2.2. Ethics

The study procedures were carried out in accordance with the Declaration of Helsinki. The Institutional Review Board of the Shanghai University of Sport approved the study. All subjects were informed about the study and provided informed consent.

2.3. Aerobic exercise intervention

Participants in the control group maintained their normal daily routines at the drug rehabilitation bureau under staff supervision. Others in the two exercise groups were engaged in exercise programs that involved three 40-min sessions per week of aerobic exercise at moderate and high intensity, respectively, on a cycle ergometer for 12 weeks. The exercise protocol was created in accordance with the recommendations of the American College of Sports Medicine (American College of Sports Medicine, 2014). Each session of the program involved stretching of the large muscle groups as warm-up (5 min), aerobic exercise as the main exercise stage (30 min), and stretching of the center-based muscles as cool-down (5 min). The moderate-intensity group needed to reach 65–75% of the indirect maximum heart rate (HRmax) during exercise, and the high-intensity group needed to reach 75–85% of the HRmax. HRmax values were calculated as $206.9 - 0.67 \times \text{age}$. Participants' HRs were monitored throughout the entire procedure using Smart Sensors (Suunto, Vantaa, Finland). All interventions were performed at the drug rehabilitation bureau.

2.4. Working memory assessment

A classic 2-back task was used to measure working memory with E-prime 2.0 (Psychology Software Tools, Pittsburgh, PA, USA). Stimuli were the numerals 0–9, and the task consisted of 6 blocks of 14 trials each. At the beginning of each block, a white cross was displayed at the center of a black screen for 10,000 ms. In each trial, a white stimulus number was first presented at the center of the screen for 500 ms. The participant was asked to match the stimulus number identity between the current stimulus and the probe number within each 2-back sequence. They were required to press one of two buttons to indicate a match or mismatch response within 2500 ms after the current stimulus had disappeared, and then proceeded immediately to the next trial (Fig. 1). The ratio of matching to mismatching stimuli was 1:2. Four practice blocks were provided prior to the formal experiment. Participants completed a pre-test of the 2-back task before participating in the intervention, and a post-test after intervention completion (Fig. 2).

2.5. Behavioral data processing

Participants' reaction times (RTs) and discernment (d') in the 2-back task were analyzed. To minimize the impact of RT outliers, trials with RTs > 1000 ms were discarded. In addition, a trimming procedure was applied in Matlab (MathWorks Inc., Natick, MA, USA) to discard outliers falling outside three standard deviations around the mean. In

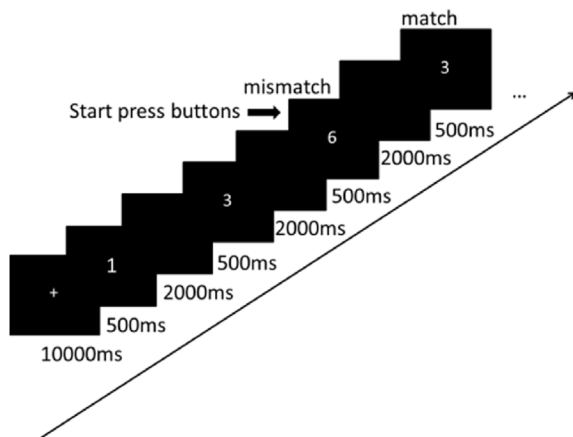


Fig. 1. Flow chart of 2-back task.

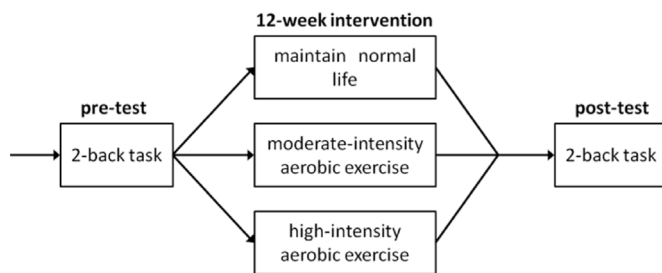


Fig. 2. Schema of current design.

accordance with signal detection theory (Swets & Green, 1972), based on the numbers of correct (e.g., responding “match” when a matching stimulus was actually present) and incorrect responses (e.g., responding “mismatch” to a mismatching stimulus), we computed hit rates and false alarms separately for all groups in pre-test and post-test. The sensitivity index, d' , indicates the degree to which a participant was able to discriminate a true signal from noise; it was computed by subtracting the z score corresponding to the false alarm rate from that corresponding to the hit rate. Sensitivity increases with the d' value.

2.6. fNIRS data acquisition and processing

A multi-channel continuous-wave fNIRS instrument (NIRScout; NIRx Medical Technologies LLC, Minneapolis, MN, USA) was used to monitor hemodynamic activity. The fNIRS probe consisted of eight dual-wavelength sources (780 and 830 nm) and seven optical detectors, with a sampling rate of 7.81 Hz. The probes were arranged according to a 10–20 system, with some adjustment to ensure that each emitter was 3 cm away from its corresponding detector. One source and one detector formed a channel, and 20 channels were formed (Fig. 3b). Based on the recent finding that the frontal cortical regions were activated robustly mainly during the n-back task (Owen, Mcmillan, Laird, & Bullmore, 2010), all channels were set in the PFC. Information on the correspondence between fNIRS channel locations and specific brain regions was taken from Okamoto et al. (2009) and Tsuzuki et al. (2007). All corresponding electrode locations were converted to standard Montreal Neurological Institute coordinates using a probabilistic estimation method (Simone, Pietro, & Marco, 2011). The locations of all channels in the PFC were estimated (Table 2), and six regions of interest (ROIs) were formed.

Data were analyzed using Homer2 open source software (MGH-Martinos Center for Biomedical Imaging, Boston, MA, USA), implemented in Matlab. Motion artifacts were detected and removed using channel-based cubic spline interpolation in Homer2 (Scholkmann, Sipchitig, Muehleemann, & Wolf, 2010). fNIRS signals were preprocessed

with a high-pass filter using the cut-off frequency of 0.04 Hz to remove baseline drift, and a 0.5-Hz low-pass filter to reduce the impacts of heartbeat pulsations, blood pressure, and skin blood flow on the signal. The modified Beer-Lambert law was applied to calculate hemoglobin concentration changes. The data were block averaged from the time window 0–30 s to obtain an average response to the 2-back task for each of the six ROIs in each participant for the pre-test and post-test. The concentration of the oxygenated hemoglobin ([HbO]) and deoxygenated hemoglobin signals ([Hb]) were chosen as the primary metrics.

2.7. Electroencephalographic data recording and processing

Continuous electroencephalographic (EEG) data were recorded from a 32-channel amplifier using the Brain Vision Recorder software (Brain Products, Munich, Germany). Due to the fixed optode positions, the electrodes were embedded in a combined fNIRS–EEG cap according to the 10–20 system (Fig. 3a). Additionally, electrooculography was performed to measure AF1 and AF2 for the detection of eye movement artifacts. The ground electrode and the online reference electrode were placed at CPz and FCz, respectively. Electrode impedance was kept below 10 k Ω . With a sampling rate of 500 Hz, the signal was bandpass filtered online at 0.1–100 Hz.

Offline data from the EEG recordings were analyzed using the Vision Analyzer 2.0 system (Brain Products, Gilching, Germany). Epochs were then filtered with an IIR filter (IIR = infinite impulse response; low cut-off, 0.1 Hz; high cut-off, 30 Hz). Ocular correction of the filtered data was conducted using the Gratton and Coles method (Gratton, Coles, & Donchin, 1983). EEG recordings were then segmented into trials from 200 ms before cue onset to 1000 ms after target onset. Trials were rejected when the voltage exceeded ± 80 μ V. ERP data were then baseline corrected to the mean amplitude of the pre-stimulus interval.

Based on the participants' grand-average waveforms and the findings of a recent study (Yuan et al., 2016), we analyzed two components that occurred before behavioral responses, N1 and P2. Therefore, the mean amplitude of the N1 component was measured at P7 and P8 (time window, 180–200 ms). The mean amplitude of the P2 component was measured at FC1 and FC2 (time window, 195–200 ms).

2.8. Statistical analysis

The assumptions of normal distribution were met. The effects of the experimental conditions on behavioral (RT and d'), fNIRS ([HbO] in ROIs), and ERP (N1 and P2 amplitudes) data were determined by a 2 (test time: pre-test, post-test) $\times$ 3 (group: control, moderate-intensity exercise, high-intensity exercise) mixed-design analyses of variance (ANOVAs) using SPSS 20.0 software (SPSS Inc., Chicago, IL, USA). When a main effect or interaction was detected, the paired t -test was used for further comparison (corrected using the Benjamini–Hochberg method). To further assess the relationships between neurobiological changes and working memory processes, Pearson's correlation analyses were conducted to detect correlations of behavioral performance parameters (d' and RT) with significant changes in mean [HbO] of ROIs and ERP amplitudes (N1 and P2).

The effect size was represented by partial η^2 values, and mean values for the behavioral variables were reported with standard errors. For all statistical analyses, the alpha significance level was set at 0.05. Marginal significance was reflected by an alpha level of 0.10.

3. Results

3.1. Effects of long-term aerobic exercise on working memory performance

Two-way ANOVA revealed significant interaction between the test time and group for d' ($F_{2,42} = 3.91$, $p = 0.028$, $\eta_p^2 = 0.16$). Further

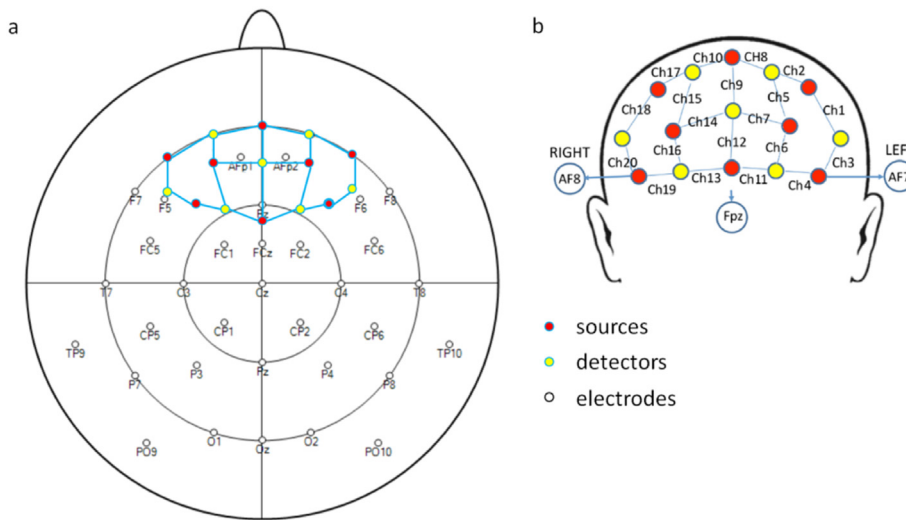


Fig. 3. (a) fNIRS optodes (red = source, blue = detector) and EEG electrodes (white) mapped on the brain. (b) fNIRS channels mapped on the brain. One source and one detector formed a channel (blue lines), and 20 channels were formed (numbered in black, i.e., Ch1). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Table 2
Location for each channel.

PFC areas	Hemisphere	Channels	MNI coordinates		
			X	Y	Z
Ventrolateral Prefrontal Cortex	L	1, 3, 4,	-41	52	-6
	R	18, 19, 20	41	51	-7
Dorsolateral Prefrontal Cortex	L	2, 5, 7, 8,	-20	45	43
	R	9, 10, 14,	19	49	42
		15, 17			
Front Polar Area		6, 12, 16	0	65	25
Orbital Frontal Cortex		11, 13	0	69	-1

L: left; R: right.

MNI: Montreal Neurological Institute

analysis showed that d' was significantly higher for the post-test than for the pre-test in the high-intensity exercise group ($p = 0.014$; Fig. 4a).

We also detected a significant main effect of the test time on RTs in the 2-back task ($F_{1,42} = 5.08$, $p = 0.030$, $\eta_p^2 = 0.11$); RTs were significantly shorter in the post-test than in the pre-test. Paired t -tests for the study groups revealed that RTs were significantly shorter in the post-test than in the pre-test for the high-intensity exercise group ($p = 0.016$), with no significant difference in the other two groups (Fig. 4b).

3.2. Effects of long-term aerobic exercise on fNIRS parameters

3.2.1. HbO

The analysis of mean [HbO] values revealed significant interaction between the test time and group ($F_{2,42} = 3.56$, $p = 0.037$, $\eta_p^2 = 0.15$) in

the left dorsolateral prefrontal cortex (DLPFC_L). Further analysis showed significantly greater activation of the DLPFC_L in the post-test than in the pre-test only in the high-intensity exercise group ($p = 0.025$; Fig. 5a). There were no other main effects or interactions.

3.2.2. Hb

The analysis of mean [Hb] values revealed a marginally main effect of group ($F_{2,42} = 3.19$, $p = 0.052$, $\eta_p^2 = 0.13$) in the DLPFC_L; mean [Hb] of this ROI in the moderate-intensity group was obviously greater than control group ($p = 0.049$; Fig. 5b).

Also, there was a marginally significant interaction between the test time and group ($F_{2,42} = 2.64$, $p = 0.083$, $\eta_p^2 = 0.11$) in the DLPFC_L. Further analysis showed significantly increased mean [Hb] of the DLPFC_L in the post-test than in the pre-test only in the control group ($p = 0.038$; Fig. 5b).

Additionally, there was a significant main effect of group ($F_{2,42} = 4.74$, $p = 0.014$, $\eta_p^2 = 0.18$) in the OFC; mean [Hb] of this ROI in the moderate-intensity group was obviously greater than control group ($p = 0.013$; Fig. 5c).

There were no other main effects or interactions.

3.3. Effects of long-term aerobic exercise on ERP data

One subject from the control group was excluded because of excessive artifacts; ERP data from the remaining 44 participants were included in the final analyses.

3.3.1. N1

ANOVA showed significant interaction between the test time and group on the mean amplitude of the N1 component ($F_{2,41} = 4.08$,

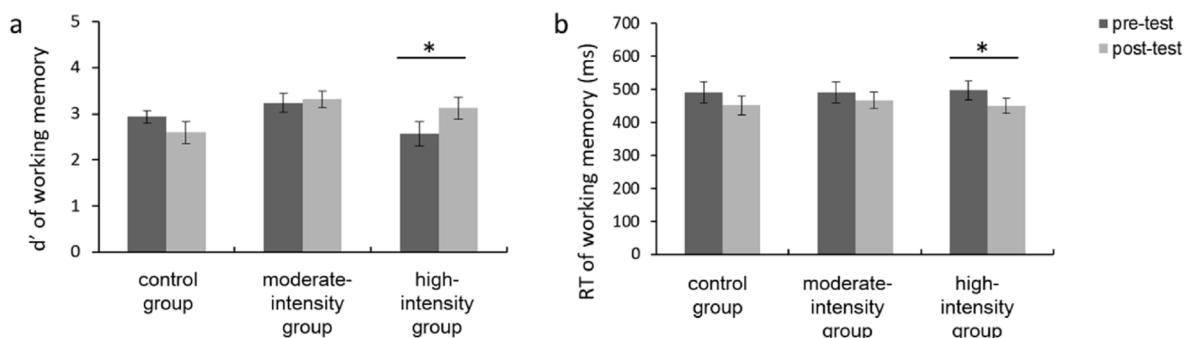


Fig. 4. RTs (a), and d' values (b) on the 2-back task. * $p < 0.05$.

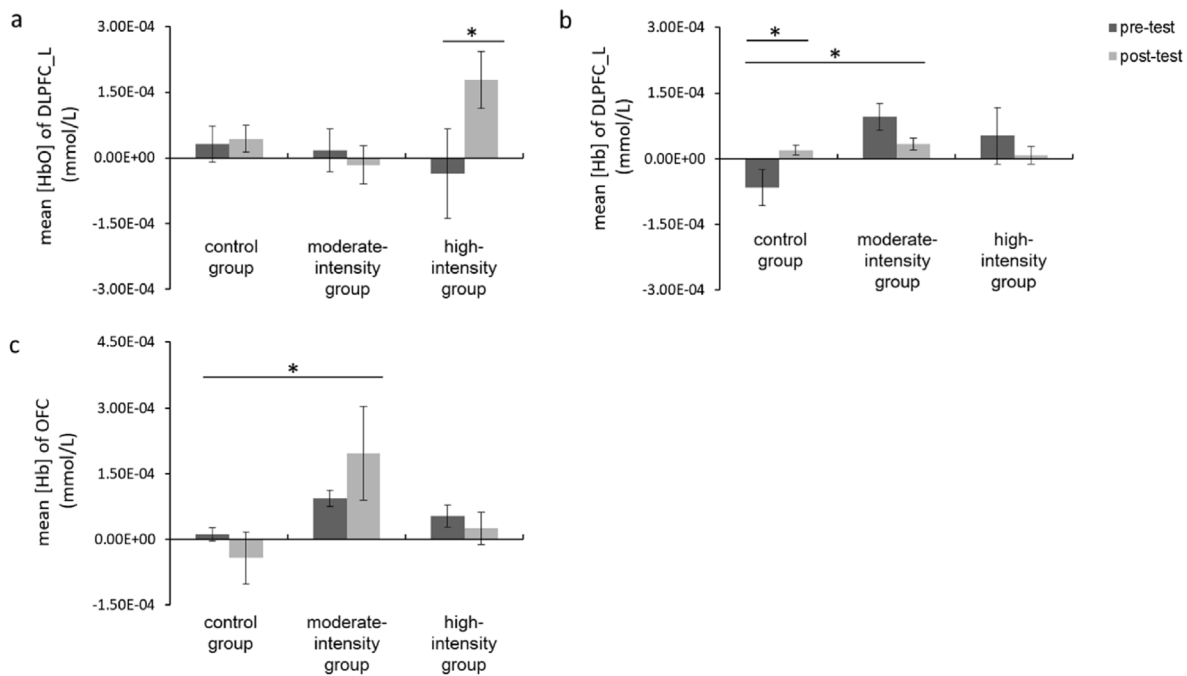


Fig. 5. Mean [HbO] values (a), mean [Hb] values (b) of DLPFC_L, and mean [Hb] values of OFC (c) on the 2-back task. * $p < 0.05$.

$p = 0.024$, $\eta_p^2 = 0.17$). Further analysis showed that the N1 amplitude was significantly decreased in the post-test relative to the pre-test only in the high-intensity exercise group ($p = 0.020$; Fig. 6a).

3.3.2. P2

ANOVA showed a significant main effect of the test time on the mean amplitude of the P2 component ($F_{1,41} = 4.07$, $p = 0.050$, $\eta_p^2 = 0.090$); the P2 amplitude was significantly decreased in the post-test relative to the pre-test. Paired t -tests for the study groups revealed that the P2 amplitude during the post-test was marginally reduced relative to that during the pre-test in the high-intensity exercise group ($p = 0.051$), with no significant difference in the other two groups (Fig. 6b).

3.4. Correlations between behavioral and neurobiological data

The analysis of correlations between behavioral data and hemodynamic responses in DLPFC_L showed that the mean [HbO] correlated marginally with d' ($r = 0.20$, $p = 0.055$, $n = 90$; Fig. 7a). For ERP data, the mean amplitude of the P2 component correlated negatively with d' ($r = -0.22$, $p = 0.038$, $n = 88$; Fig. 7b). No other significant correlation between behavioral and neurobiological data was observed.

4. Discussion

The purpose of this study was to determine the effects of aerobic exercise at various intensities on working memory among MA-dependent patients. Meanwhile, fNIRS and ERP data were obtained during working memory task performance to explore neurobiological changes in the brain. Broadly, the findings confirmed our hypothesis from both behavioral and neural level. A 12-week high-intensity aerobic exercise had a more positive impact than did moderate-intensity exercise on working memory in MA-dependent patients.

The behavioral data showed that the d' of working memory was improved significantly and RTs were effectively reduced, representing increased sensitivity and reaction speed, in the high-intensity exercise group alone. This finding is consistent with that of Jeon and Ha (Jeon & Ha, 2017), who found that working memory performance increased significantly in a healthy high-intensity exercise group compared with

low- and moderate-intensity groups. A previous study demonstrated that high-intensity exercise (e.g., that generating 80% HRmax) had effects on the performance of more complex memory tasks, such as three digit addition and the backward digit span (Kim, Park, & Kim, 2010). 2-back task is a working memory task with a certain complexity; hence, high-intensity exercise resulted in stronger performance improvement. Our hypothesis was thus confirmed at the behavioral level.

Importantly, a combined fNIRS-ERP measure was used to identify the neurofunctional underpinnings of changes in working memory processes in this study. Our fNIRS data showed that the mean [HbO] in the DLPFC_L during 2-back task performance was significantly higher after the high-intensity exercise intervention. Meanwhile, the mean [Hb] in this ROI was obviously increased only in the control group during post-test. Those findings revealed that long-term high-intensity aerobic exercise could have a positive effect on concentration levels related to hemodynamic responses in the DLPFC_L, while participants without exercise intervention would experience a decline in the activity of this area. The DLPFC is known to be involved in executive function, an umbrella term for the management of cognitive processes, including working memory (Barbey, Michael, & Jordan, 2013), and reduced activity in this area correlates with poor performance on working memory tasks. Our results are consequent to existing studies that aerobic exercise enhances prefrontal-dependent aspects of cognitive performance, such as working memory (Jeon & Ha, 2017; Moreau, Kirk, & Waldie, 2017), and induces physiological changes in brain regions, including changes in brain structure (Gondoh et al., 2009), cerebral blood flow (Lista & Sorrentino, 2010), and BDNF levels (Jeon & Ha, 2017). Moreover, the change in cerebral blood flow increases with the intensity of exercise (Subudhi, Miramon, & Roach, 2009), inducing more brain activation and enhancing neural function. This finding confirmed our hypothesis that the effects of high-intensity exercise on working memory are stronger than those of moderate-intensity exercise at the neural level. Additionally, post-test fNIRS data from the high-intensity exercise group showed significant [HbO] increases and [Hb] decreases relative to pre-test levels, indicating that the exercise intervention reduced Hb levels via increases in blood flow that overcompensated for increased DLPFC_L oxygen consumption.

Furthermore, if changes in oxygen consumption and cerebral blood flow in the brain are specific to working memory tasks, one may assume

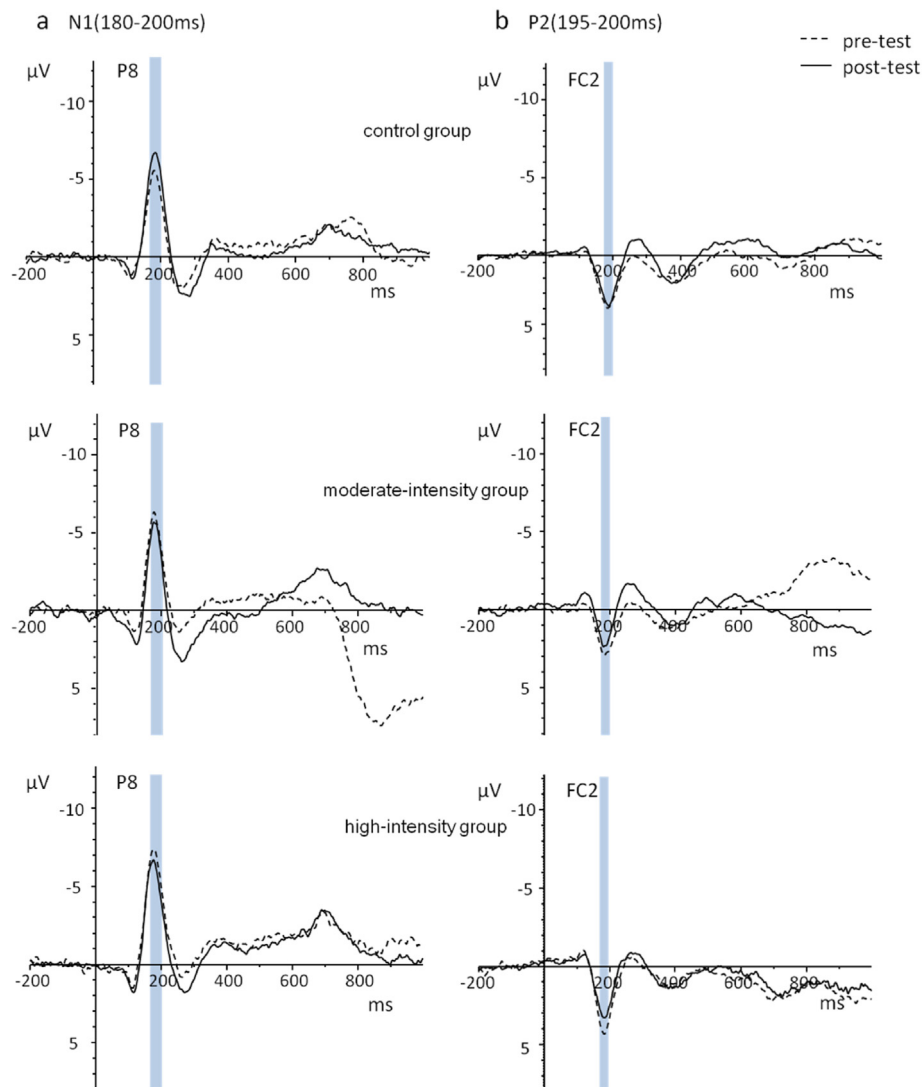


Fig. 6. Grand average ERPs at electrodes FC2 and P8 elicited during 2-back task performance. The gray areas highlight the time windows for N1 (180–200 ms) and P2 (195–200 ms) that were used for the statistical analysis.

that hemodynamic responses are related to behavior (Schroeter, Zysset, Kupka, Kruggel, & Von Cramon, 2002). In this study, the mean [HbO] in the DLPFC_L was weakly and positively related to d' for the 2-back task. Since the intervention was not long enough, this correlation was a tad weak, but we observed a significant trend of association between neural activity in the DLPFC_L and behavioral working memory performance. PFC activity appears to be an important part of the effect of aerobic exercise on working memory processes; future research should

be conducted to validate the present findings and further advance knowledge in this area of research.

The mean [Hb] of the OFC was greater in the moderate-intensity exercise group than in the control group, which showed a decline in OFC activity. Barbey, Koenigs, and Grafman (2010) provided neuropsychological evidence for the mandatory involvement of the OFC in executive control functions underlying joint working-memory maintenance, manipulation, and monitoring processes; decreased activity in

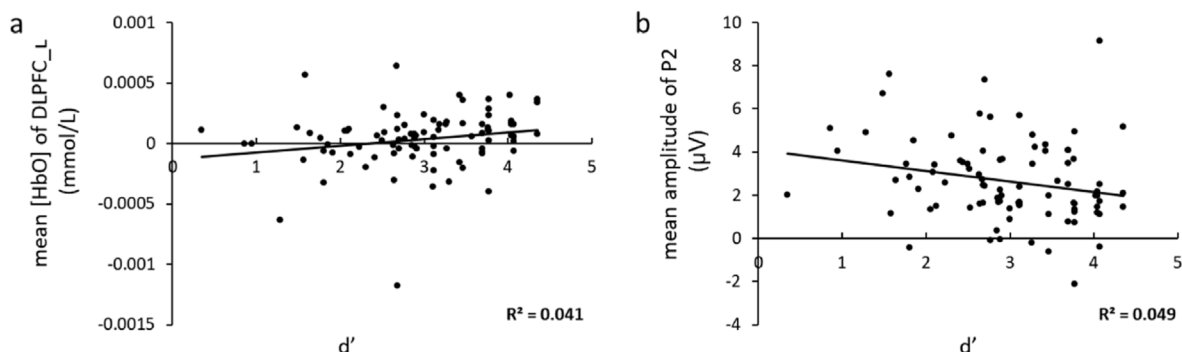


Fig. 7. Correlations of d' with the mean [HbO] in the DLPFC_L (a) and mean amplitude of P2 (b).

this area compromised *n*-back task performance. In contrast, the current study revealed no significant difference in behavioral findings among the three groups, perhaps due to differences in the involvement of the OFC in working memory between healthy and MA-dependent participants.

Synchronized electrophysiological data showed similar findings. As expected, ERP results obtained in this study indicated that aerobic exercise affected the dynamic time course of neural activity during working memory processing. Lower N1 amplitudes were found only in the high-intensity exercise group. The N1 component is related to the early processing of stimuli and is sensitive to the physical properties of the stimuli (Morgan, Klein, Boehm, Shapiro, & Linden, 2008). Moreover, some researchers have found that the N1 component, as an index of discrimination processing, reflects operations within attention focusing (Vogel & Luck, 2000). The lower N1 amplitude observed in this study indicates that participants could identify whether stimuli matched with less use of mental resources after the intervention. P2 amplitudes were also lower in the high-intensity exercise group. Previous studies have shown that the P2 component indicates the initial stage of context updating in working memory (Lenartowicz et al., 2010), which reflects an early stage of information processing, a core stage in many cognitive tasks, including the *n*-back task (Ciecko-Michalska et al., 2012). This early stage of updating is a crucial step for successful task performance (Yuan et al., 2016). Our results are consistent with this viewpoint, given the significant correlation of behavioral data with the P2 amplitude. Evidence in the literature also suggests that the P2 component is associated with the direction of attentional resources to stimuli (Yang, Guan, Dedovic, Qi, & Zhang, 2012). Thus, MA-dependent patients could better direct their attention to positive signals in the early processing stage after the 12-week high-intensity exercise intervention, implying that they could better control their attention to drug rewards and reduce drug cravings.

Several limitations of this study point to the need for cautious interpretation of the results and directions for further exploration. Although we used the HRmax formula recommended by the American College of Sports Medicine (2014) and used widely in similar studies (Wang et al., 2016, 2017) to define exercise intensities, HRmax may evolve during long-term exercise. Additionally, we only monitored participants' HR data to control exercise intensity; we did not record them. As a consequence, the exercise intensity may vary considerably with the use of different intervention periods. In future studies, HR values should be recorded throughout the intervention period, which can restore exercise performance in a fatigued state (Brown & Bray, 2019). The exercise intensity classification should be improved through the use of a more stable variable, such as the critical power concept (Jones & Vanhatalo, 2017; Jones, Vanhatalo, Burnley, Morton, & Poole, 2010). In addition, baseline cognitive data should be collected before the intervention to understand the degree of existing cognitive damage in MA-dependent patients.

5. Conclusion

The current study aimed to investigate the beneficial effects of aerobic exercise on working memory in MA-dependent patients, with the use ERPs and fNIRS to detect underlying neural processes. The results suggest that MA-dependent patients discriminated more sensitively during working memory processes after a high-intensity exercise intervention. On the neural level, high-intensity exercise significantly enhanced neural activity in the DLPFC-L and improved attention control, as shown by lower amplitudes in the N1 and P2 components. These results indicate that high-intensity exercise has an all-encompassing cognitive benefit, and highlight it as a promising protective factor against cognitive decline. Given this evidence of the relationship between high-intensity exercise and working memory, the authorities should advocate regular participation in high-intensity exercise as a therapeutic tool for MA-dependent patients. Different intensity

exercises bring different exercise effects on working memory suggesting that this relationship needs to be studied in further detail using objective measures of intensity with MA-dependent patients.

Ethics

The study procedures were carried out in accordance with the Declaration of Helsinki. The Institutional Review Board of the Shanghai University of Sport approved the study. All subjects were informed about the study and all provided informed consent.

Role of funding source

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CRediT authorship contribution statement

Yifan Chen: Writing - original draft. **Yingzhi Lu:** Writing - review & editing. **Chenglin Zhou:** Funding acquisition. **Xiaochun Wang:** Writing - review & editing.

Declaration of competing interest

The authors declare no conflict of interest.

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